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Application Note: The compatibility between Vortex86SX and Vortex86DX

3rd release: 2009/4/27

Please ignore the 1st version (2008/06/13) and 2nd version (2008/6/25)

Mainly this 3rd release is to clarify Vortex86DX version and the corresponding date code which printed on the chips

Subject:

1, The pin definition difference between Vortex86SX and Vortex86DX, and reference schematic.

2, Pin location change between Vortex86DX , version ID: A9120 and version ID: A9121and recommended layout.

The difference between A9120 and A9121 are as below.

- **Enhance the IDE compatibility, change from ITE821X (A9120) to general IDE (A9121).**
- **DQS pins, pin A7 and D14 shifted, and support the 2nd bank of its 16-bit DDR2 Bus on A9121.**

Code change from 0xxx-x-PB (A9120) to 0xxx-x-PBGA (A9121). The easiest to recognize the version is see if it's PB or PBGA.

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1, The pin definition difference between Vortex86SX and Vortex86DX, and reference schematic.

In order to have Vortex86DX support higher DDRII clock, the Vortex86DX must have ODT (on-die signal termination) to improve the DDRII Differential clock inputs. The pin definition difference between Vortex86SX and Vortex86DX is as below.

Ball No#	Vortex86SX	Vortex86DX version **PB**	Vortex86DX version **PBGA**	Note
A3	NC	ODT[1]	ODT[1]	
A5	NC	VREF_IN	VREF_IN	
A6	NC	DQS_[0-]	DQS_[0-]	
A7	NC	DQS_**[1-]**	**DQS_[0+]**	
D14	DQS0	DQS_**[0+]**	**DQS_[1-]**	
E16	DQS1	DQS_[1+]	DQS_[1+]	
B3	NC	ODT[0]	ODT[0]	
B4	NC	CS_1	CS_1	
B5	NC	VREF_OUT	VREF_OUT	
B6	NC	OCDBIAS	OCDBIAS	1.8V 200ohm 1/%
B13	CS_1	MA14	MA14	
C3	NC	CKE	CKE	
D1	NC	DP	DP	
D9	VDLL1	NC	NC	
D10	VDLL0	NC	NC	
E1	NC	DM	DM	
E9	GNDDLL1	NC	NC	
E10	GNDDLL0	NC	NC	
J3	TESTCLK	NC	NC	
J4	VPLL0	AVDD18_1	AVDD18_1	VPLL0: 3.3V, AVDD18_1:1.8V
H5	GNDPLL0	AVSS18_1	AVSS18_1	
H4	VPLL1	AVDD18_2	AVDD18_2	VPLL1: 3.3V, AVDD18_2:1.8V
J5	GNDPLL1	AVSS18_2	AVSS18_2	
AA24	Vdd_pll_1	NC	NC	
Y24	Vss_pll_1	NC	NC	

2, Pin location change between Vortex86DX , version ID: A9120 and version ID: A9121and recommended layout.

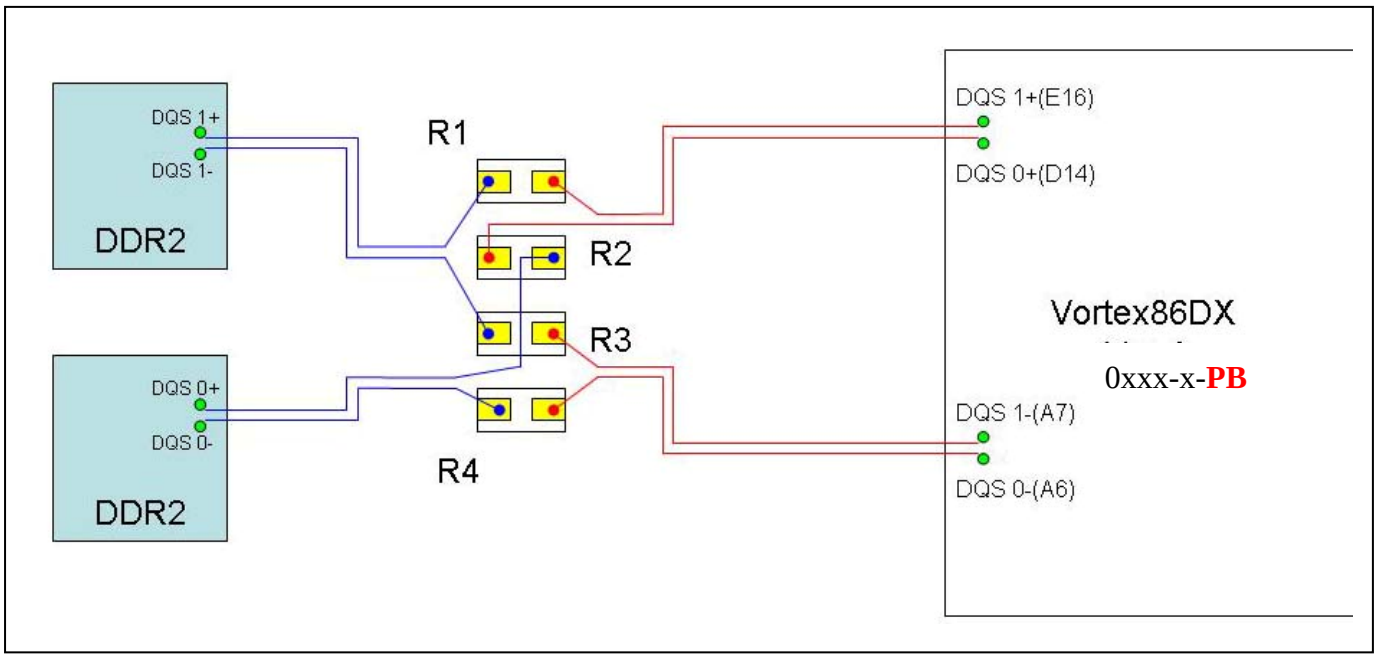
To improved the signal integrity of data strobe DQS, the same channel of differential pair DQS should be near-by, DQS0+ & DQS0- , DQS1+ & DQS1- .

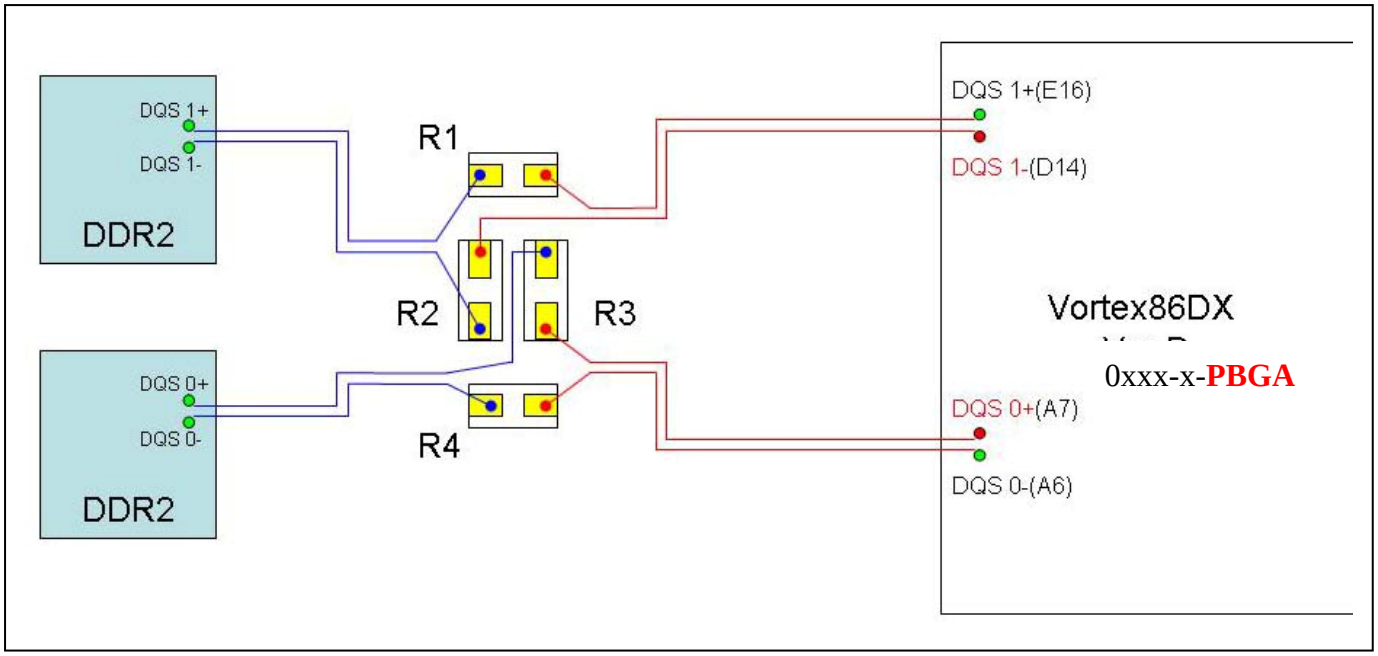
In the current version ID: A9120 of Vortex86DX (code 0xxx-x-**PB**). The DQS0+ (Ball # D14) and DQS1+ (Ball # E16) , DQS0- (Ball # A6) and DQS1- (Ball # A7) are near-by, which need to be improved.

In the version ID: A9121 of Vortex86DX (code 0xxx-x-**PBGA**). The DQS1- and DQS0+ will be switched. Which the DQS1- will move to ball # D14, DQS0+ will move to ball # A7.

To minimize the engineering change between version A and B. Please see the recommended layout as below.

---------- see next page ----------





When you layout the Vortex86DX board, please have the R2 and R3 close to each other and with the properly interval and pad size (depends on your resistor size). When migrate from A9120 **(PB)** to A9121 **(PBGA)** version, please populate the resistors on R2 and R3 with correctly direction (rotate 90 degree).

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